

Rigid Motions on the Coordinate Plane

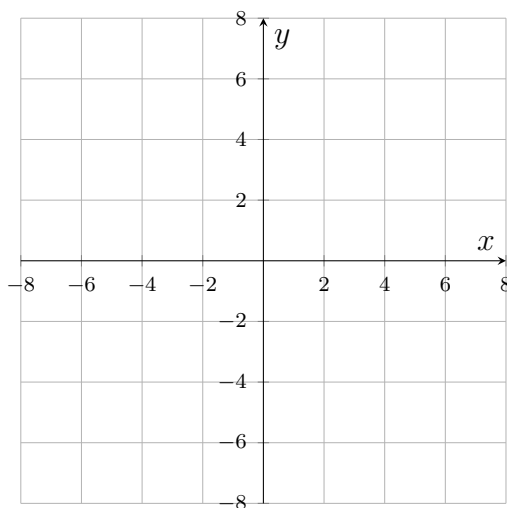
Every slide, flip and turn written as a rule on coordinates — and every one of them keeps distances the same

Motion	Rule	Motion	Rule
Translation	$(x + a, y + b)$	Reflect in $y = x$	(y, x)
Reflect in x -axis	$(x, -y)$	Reflect in $y = -x$	$(-y, -x)$
Reflect in y -axis	$(-x, y)$	Reflect in $x = c$	$(2c - x, y)$
Half turn, origin	$(-x, -y)$	Quarter turn ccw, origin	$(-y, x)$

Quarter turn counterclockwise about (h, k) : $(x, y) \rightarrow (h - (y - k), k + (x - h))$

- Read the rule, substitute, simplify — then check the answer against the picture in your head.
- Distance between points: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

1. Point $A(-4, 1)$ is translated by the rule $(x, y) \rightarrow (x + 6, y - 3)$. Plot A and its image on the grid, then write the coordinates of A' .



$A' (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

2. Point $B(5, 2)$ is reflected in the line $y = -x$. Write the coordinates of B' .

B' (_____, _____)

3. Point $C(-6, 3)$ is given a half turn about the origin. Write the coordinates of C' .

C' (_____, _____)

4. Segment $\overline{PQ}$ has endpoints $P(-3, 2)$ and $Q(1, 5)$. The whole segment is translated by $(x, y) \rightarrow (x + 4, y - 6)$, giving $\overline{P'Q'}$.

(a) Write the coordinates of P' . P' (_____, _____)

(b) Find the length PQ . _____

(c) The image of Q is $Q'(5, -1)$. Find the length $P'Q'$. _____

5. Point $D(9, 4)$ is reflected in the vertical line $x = 2$. A reflection in a vertical line leaves the y -coordinate alone, so D' has y -coordinate 4.

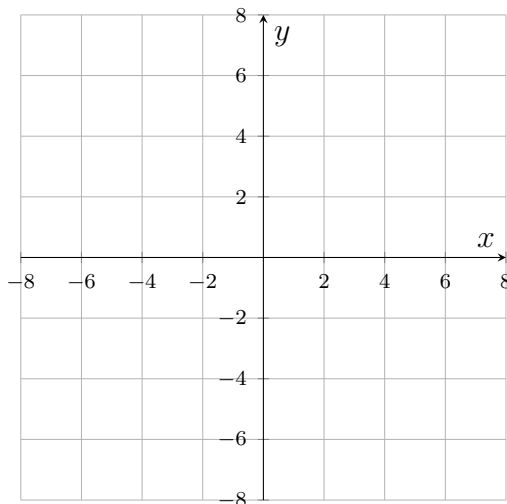
(a) Find the x -coordinate of D' . _____

(b) Find the length DD' . _____

6. Point $E(3, -7)$ is turned a quarter turn counterclockwise about the point $(1, 2)$. Write the coordinates of E' .

E' (_____, _____)

7. Triangle ABC has $A(2, 1)$, $B(6, 1)$ and $C(6, 4)$. It is first reflected in the y -axis, and that image is then translated by $(x, y) \rightarrow (x, y + 5)$. Plot the triangle on the grid if it helps you see the two steps.



(a) Write the coordinates of A'' , the final image of A . A'' (_____, _____)

(b) Find the length AB . _____

(c) Explain why $A''B''$ must have the same length as AB , without computing B'' .

8. Segment $\overline{EF}$ has endpoints $E(-5, -2)$ and $F(3, 4)$.

(a) Find the length EF . _____

(b) E is given a half turn about the origin. Write the coordinates of E' . E' (_____, _____)

(c) The same half turn sends F to $F'(-3, -4)$. Find the length $E'F'$. _____

(d) Explain why every rigid motion has to leave the length of $\overline{EF}$ unchanged.